Student's Index No.: _____ Signature: _____

UNIVERSITY OF GHANA

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DEPARTMENT OF TEACHER EDUCATION

SCHOOL OF EDUCATION AND LEADERSHIP

COLLEGES OF EDUCATION

END OF SEMESTER TWO EXAMINATIONS, 2024/2025

TIME: 2 HOURS

COURSE CODE: **TEEG 106/TEJS 106/TEUP 108**

COURSE TITLE: **INTRODUCTION TO INTEGRATED SCIENCE II**

INSTRUCTIONS: This paper consists of **Sections A and B**. Answer all the questions in Section A and three from Section B.

SECTION A

[25 marks]

Answer all the questions in this section.

1. Which of the following options best explains the **law of conservation of energy**?
   A. Energy is constantly being created and destroyed
   B. Energy is transferred into matter
   C. Energy can only change from one form to another without loss
   D. Energy can only exist in potential or kinetic form

2. What energy is most directly responsible for a mango hanging on a tree?
   A. Potential
   B. Chemical
   C. Electrical
   D. Kinetic

3. Which of the following sources of energy is **renewable**?
   A. Diesel
   B. Coal
   C. Wind
   D. Natural gas

4. Which of the following household items is most likely to use both an electrical heating element and electronic control circuits?
   A. Incandescent light bulb
   B. Microwave oven
   C. Blender
   D. Electric drill

5. Which of the following statements is a **misconception** about energy?
   A. Energy can change forms
   B. Non-moving objects can have potential energy
   C. Energy is a substance
   D. Energy is needed to create force.

6. What are the energy transformations which occur when the **sun shines**?
   A. Electrical to thermal
   B. Nuclear to light and heat
   C. Chemical to heat
   D. Kinetic to potential

7. What state change occurs when **energy is added to water at room temperature**?
   A. Freezing
   B. Sublimation
   C. Vaporization
   D. Deposition

8. What does the "C" in the **ABC of First Aid** stand for?
   A. Care
   B. Compression
   C. Circulation
   D. CPR

9. What is the **correct depth for adult chest compressions** during CPR?
   A. 3–4 cm
   B. 4–5 cm
   C. 5–6 cm
   D. 6–7 cm

10. Which of the following actions should NOT be taken when providing first aid to a casualty with **a large burn**?
    A. Call for an ambulance
    B. Immerse the burn in cold water
    C. Remove jewelry from the affected area
    D. Check for signs of circulation

11. What is the first step in helping a **choking infant under 1 year old**?
   A. Give abdominal thrusts
   B. Start CPR immediately
   C. Administer back slaps
   D. Give 5 rescue breaths

12. Which item in the first aid kit is correctly matched with its **use**?
   A. Tweezers – controlling bleeding
   B. Pocket mask – for CPR
   C. Antiseptic wipes – removing splinters
   D. Vinyl gloves – cooling burns

13. Which of the following statements is a **major myth** about CPR and sudden cardiac arrest?
   A. Rescue breaths are unnecessary
   B. Chest compressions increase circulation
   C. CPR helps deliver oxygen to organs
   D. AED use improves survival

14. Which of the following diseases is **NOT** an example of a contagious disease?
   A. Measles
   B. Syphilis
   C. Tetanus
   D. Chicken pox

15. Which of these statements best describes an **asymptomatic carrier**?
   A. A person infected with a disease and showing symptoms
   B. A person who spreads disease but never becomes infected
   C. A person infected with a disease but not showing symptoms
   D. A person who spreads disease through animal vectors

16. Which of the following resources is non-renewable?
   A. Forest
   B. Soil
   C. Petroleum
   D. Wildlife

17. Which greenhouse gas is *directly* produced in large amounts by motor vehicle exhaust?
   A. Sulphur dioxide
   B. Carbon dioxide
   C. Methane
   D. Nitrous oxide

18. The incubation period of a disease refers to the time _____.
    A. between infection and the onset of symptoms
    B. taken to treat a disease
    C. taken for an epidemic to spread
    D. for pathogens to leave the body

19. Which of the following activities is a result of **acid rain**?
    A. Melting of icebergs
    B. Increase in atmospheric humidity
    C. Corrosion of metals and buildings
    D. Decrease in greenhouse gases

20. The practice of replacing harvested trees with seedlings in plantations is called _____.
    A. afforestation
    B. selective logging
    C. reforestation
    D. conservation cropping

21. An example of **fomite transmission** is _____.

22. The human activity that directly increases the concentration of greenhouse gases in the atmosphere is _____.

23. In a simple electronic circuit, the component that consumes power to perform a particular function is called the _____.

24. The application of scientific knowledge to produce tools, machines and medicines for human use is referred to as _____.

25. The compilation of the students' work over a period of time that shows developmental progress is known as is _____.

SECTION B

[75 marks]

Answer three (3) questions from this section.

QUESTIONS			MARKS
1.	a.	Explain the following forms of energy.	
	(i)	Heat energy	3
	(ii)	Potential energy	3
	(iii)	Kinetic energy	3
	(iv)	Chemical energy	3
	b. (i)	Sherifatu lifts a body of mass 75000g to a height of 6m. Find the potential energy of the body. [g = 10ms-2].	3
	b. (ii)	What are contagious diseases?	2
	c.	State whether the diseases below are viral or bacterial diseases.	7
	(i)	Cholera	
	(ii)	Measles	
	(iii)	Whooping cough	
	(iv)	Tetanus	
2.	a. (i)	Describe the **ABC of First Aid.**	3
	a. (ii)	Explain why **each** step in **a (i)** is critical during an emergency.	6
	b.	Explain, step by step, how to perform **CPR** on an **adult who is unresponsive and not breathing**. Your response should include compressions and rescue breaths.	10
	c.	Explain any **three** (3) reasons why a trainee teacher must keep a portfolio.	6
3.	a.	Discuss **five** (5) methods of forest conservation and explain how each helps to protect the environment.	10
	b.	Explain the following terms in terms of disease outbreak:	
	(i)	Endemic	2
	(ii)	Pandemic	2
	(iii)	Epidemic	2
	(iv)	Pathogenic disease	2
	(v)	Zoonotic disease	2
	c.	Give **two** (2) examples of each of the following categories of disease:	
	(i)	Zoonotic disease	2
	(ii)	Non-infectious disease.	2
	d. (i)	What does the acronym PVA in the National Teachers' Standard stand for?	1
4.	a. (i)	Explain **three** (3) types of pollution.	6

	a. (ii)	Discuss **two** (2) effects of each type of pollution mentioned on the environment and human health.	4
	b.	Discuss **four** (4) benefits of forest reserves and game reserves to humans and the environment.	8
	c.	Explain **three** (3) assessment techniques you would use to evaluate learners' understanding of forest conservation.	6
5.	a.	Consider the following diseases: **typhoid**, **tetanus**, **diphtheria, pneumonia,** and **gonorrhea**.	
	(i)	Name the bacterium responsible for each disease.	5
	(ii)	Give **two** (2) symptoms of each disease.	8
	(iii)	State any treatment measure or way of prevention of each disease.	4
	b.	Explain **two** (2) consequences of global warming.	4
	c.	As a professional science teacher, name any 5 items you would include in your teaching portfolio.	5

End of paper